

What are the corresponding reported percent missingness of lipid variables in NHANES

Reported missingness of lipid variables in NHANES

The included papers largely analyze NHANES lipid data but almost never report detailed **percent missingness by lipid variable**. Only one paper provides explicit overall missingness percentages for standard lipids.

Explicitly reported missingness

Overall lipid-panel missingness in NHANES 1999–2018

Gao et al. (US trends in cholesterol screening and lipids) report, for their NHANES analytic sample (1999–2018) (Gao et al., 2023):

Measure / variable group	Percent missing	Citations
Cholesterol screening (self-report)	3.0%	(Gao et al., 2023)
Total cholesterol (TC) + HDL-C	6.2%	(Gao et al., 2023)
Triglycerides (TG) + LDL-C	6.8%	(Gao et al., 2023)
Lipid-lowering medication use	0.1%	(Gao et al., 2023)

FIGURE 1 Reported missingness for NHANES lipid variables

These percentages are for the combined 1999–2018 cycles and reflect the share of participants with missing data on each outcome in that specific analysis, not official NHANES-wide missingness rates.

Information from other NHANES lipid analyses

Other NHANES-based studies state that participants lacking specific lipid values or lipid data were **excluded**, but they do not quantify percent missingness for each lipid variable:

- Li et al. (NHANES 2007–2018) excluded participants “with missing lipid values” but did not report percentages (Li et al., 2022).
- Sun et al. (NHANES 1999–2018) give counts of participants without TC, TG, and LDL-C data but not proportions relative to the full eligible sample (Sun et al., 2022).
- Several additional NHANES analyses (e.g., dyslipidemia, TC/HDL ratio, sarcopenia, environmental exposures) either require **complete lipid data** or mention excluding those with missing lipids, without reporting explicit missingness percentages (Huang et al., 2024; Zheng et al., 2024; Huang et al., 2024; Shi et al., 2022).

Conclusion

Within these abstracts and excerpts, the only quantified missingness for NHANES lipid variables is from Gao et al.: about 6–7% missing for TC/HDL-C and TG/LDL-C in their 1999–2018 analysis. Other NHANES lipid studies acknowledge excluding participants with missing lipid values but do not provide corresponding percent-missingness figures for LDL, TG, HDL, or total cholesterol.

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